

EMERGENCE LAB — STAGE 2 WHITE PAPER

Resonant Transfer, Environmental Forcing, Structured State, and Experimental Transition

Status: Stage 2 working white paper — scope and causal-hypothesis correction incorporated

Date: 2 September 2026

Boundary: Begins after the close of the Stage 1 circulation paper. Stage 1 remains frozen.

Separate lineage: The restored-anatomy V2 work (Mati, radials, orbitals, resonance rings, and atom-inspired rendering) remains a separate document and implementation track and is not incorporated into this Stage 2 paper.

RESEARCH CLAIM AND BOUNDARY

Framework capability: Emergence Lab Stage 2 provides a structured dynamical state-space framework for controlled comparison of perturbation-driven trajectories in coupled adaptive node systems.

Current supported Stage 2 claim: Under the explicit rules, controls, tested seeds, and frozen runtime of EL-EXP-REEL-001, allowing transfer efficiency to respond dynamically to current relational Theta state reproducibly changed how the same selective perturbation was processed relative to a frozen-efficiency control. Across the completed B/C replication series, this produced more cumulative modeled transfer and work and less cumulative modeled dissipation. This claim applies to the modeled system and normalized computational quantities only.

Immediate causal extension: Preserved or restored relational support may causally improve internal-state recovery following controlled state disruption. This is a testable hypothesis motivated by the Stage 1 replacement series; it is not established by EXP-22C or EL-EXP-REEL-001.

Required decisive comparison: Compare preserved or restored relational support against severed or randomized relational support while holding state damage, initialization, timing, topology conditions, and measurement constant. The favored interpretation must outperform simpler alternative mechanisms.

Explicit non-claim: Current results do not demonstrate consciousness, subjective experience, sentience, personhood, biological equivalence, preservation of ontological identity, literal thermodynamics, or validated prediction of arbitrary real-world systems.

Correction note: This clarification does not revise the recorded results of Stage 1, EXP-21C, EXP-22C, or EL-EXP-REEL-001. It distinguishes demonstrated capability and supported computational results from an unconfirmed causal hypothesis, and it states the experimental comparison required to test that hypothesis.

ABSTRACT

Emergence Lab Stage 2 shifts the project from a scalar/partially structured exploratory simulation lineage toward a deliberately structured, reproducible experimental system. The core node is represented as

$$N_i(t) = (X_i(t), Y_i(t), Z1_i(t), Z2_i(t), \text{Theta}_i(t)),$$

with each coordinate treated as structured state rather than as a single diagnostic scalar. The split between Z1 and Z2 is retained because failed-transfer recovery and successful integration are not the same operation. Theta carries resonance information, including frequency, phase, amplitude, and coherence.

Stage 2 also clarifies an older design lineage that predates the mature Z1/Z2 vocabulary. Early handwritten work described a resonance tuple approximately of the form

$$\text{eta} = (f, A, \text{phi}, \text{gamma}),$$

where frequency/relatedness, amplitude/strength, phase/alignment, and damping/resistance were associated with a system that could self-organize toward lower interaction cost. This is treated as conceptual ancestry of the modern resonance state, not as proof that any old symbol maps one-to-one onto a present variable.

A central Stage 2 hypothesis is that environmental perturbations can provide excitation to an open node system, while resonance similarity modulates the efficiency with which that excitation is transmitted between nodes. Energy is therefore not introduced as an additional identity coordinate or internal battery. Instead, excitation, transfer, dissipation, work, and carried flow are accounted for as derived interaction quantities. A candidate law for the first controlled experiment is

$$\begin{aligned}\text{eta}_{ij} &= S_Theta(i,j) \\ C_{ij} &= 1 - S_Theta(i,j),\end{aligned}$$

where S_Theta is pairwise resonance similarity. These equations are explicitly provisional and experimentally replaceable.

The first Stage 2 preregistered test, EL-EXP-REEL-001, asks whether selective environmental exposure has a different persistent effect when transfer is dynamically resonance-dependent than when efficiency is frozen. It uses a 2×2 factorial design with the primary causal contrast (A-B)-(C-D). The lean runtime excludes restored visual anatomy so that the immediate causal surface remains minimal.

1. STAGE BOUNDARY

Stage 1 established a useful experimental lineage but also revealed important limitations: scalar compression, ambiguous interface labels, legacy runtime artifacts, implementation drift, and

insufficient separation between conceptual architecture and actually executed mechanisms. Stage 2 does not rewrite Stage 1. Results obtained under Stage 1 remain evidence about the runtime and protocol that generated them.

Stage 2 begins when the laboratory adopts three rules:

1. the experimental state and transfer mechanisms must be explicit enough to inspect;
2. initialization and observation must be reproducible enough to support causal comparisons; and
3. proposed mechanisms must be labeled according to status rather than being silently promoted into canonical architecture.

For that reason, this paper uses three status labels:

Adopted architecture: a project-level structural decision currently required for Stage 2 work.

Candidate mechanism: an explicit rule implemented for testing but not established as a general law.

Unresolved: a question deliberately left open pending experiment or Senate decision.

2. CANONICAL STAGE 2 NODE

2.1 Adopted core state

The Stage 2 core node is:

$N_i = (X_i, Y_i, Z1_i, Z2_i, \Theta_i)$.

No standalone E coordinate is part of the core node.

X — Identity / persistent organization

X represents the node's comparatively durable present organization. In Stage 2 it is structured state. A scalar summary may be displayed for telemetry, but that projection is not the full variable.

Y — Memory / perception / retained interaction state

Y carries recent and accumulated effects of input and interaction. It is structured state and acts as an important bridge between environmental exposure and later node behavior.

Z1 — recoil / recovery / failed-transfer response

Z1 is reserved for the response associated with mismatch, failed integration, recovery, recoil, or structural restoration. It must not be silently collapsed into Z2.

Z2 — integration / transfer / carryover response

Z2 is reserved for qualified transfer, integration readiness, accepted carryover, or successful incorporation. It must not be silently collapsed into Z1. In the older anatomical lineage, the successful-transfer branch was conceived as kinetic-like: an orbital/radial carrier moving energy and memory from the Y side toward X. Stage 2 preserves that lineage as architectural context without equating Z2 to a static store of literal kinetic energy.

Theta — resonance state

Theta represents the node's resonance-bearing state. The current operational representation includes frequency, phase, amplitude, and coherence. Pairwise similarity between two Theta states is written $S_Theta(i,j)$ in $[0,1]$. A scalar Theta display value is only a projection of this richer state.

2.2 Structured rather than scalar

Stage 2 treats all five core coordinates as capable of internal structure. This is not a claim that one particular dimensionality is final. The important architectural rule is that identity, memory, recoil, transfer, and resonance are not assumed to be exhaustively represented by one number each.

The present EL-EXP-REEL-001 runtime uses three-component vectors for X, Y, Z1, Z2, while Theta contains structured frequency, phase, amplitude, and coherence. This dimensionality is an experimental implementation choice, not a metaphysical or biological claim.

3. THE Z SPLIT IS NOT OPTIONAL IN STAGE 2

Earlier project phases sometimes treated Z as a generic stability variable. Stage 2 rejects that collapse for experiments involving transfer.

A failed interaction and a successful integration are mechanically different events. If only Z2 exists, there is no explicit recovery/recoil path after mismatch. If only Z1 exists, there is no explicit pathway for accepted transfer and carryover.

The Stage 2 distinction is therefore:

mismatch / failure -> Z1 response
qualified transfer -> Z2 response

The exact differential equations governing recovery and integration remain candidate-level and may change with evidence. The existence of two distinct response channels is the adopted architectural decision.

4. RESONANCE LINEAGE AND THE EARLY LOW-COST INTERACTION IDEA

Handwritten work from the earlier project period describes a resonance-like tuple:

$\eta = (f, A, \phi, \gamma)$

with notes associating f with frequency or relatedness; A with amplitude, strength, or influence; ϕ with phase and alignment; and γ with damping, resistance to change, or stability.

The same notes describe a system in which nodes can self-organize toward stable states by minimizing interaction cost and characterize harmonious interaction as comparatively low-effort.

Stage 2 recognizes a strong conceptual continuity between this early work and the modern Theta variable. However, the historical artifact does not establish that $\gamma = Z1$, nor that $Z2$ is literally kinetic energy. Those mappings would be retrospective overreach.

The supported conceptual bridge is narrower:

better resonance alignment $\rightarrow$ lower effective mismatch/resistance $\rightarrow$ more efficient transfer.

This bridge becomes a testable Stage 2 hypothesis rather than a retrospective claim about Stage 1.

5. ENVIRONMENTAL FORCING AND THE ENERGY QUESTION

5.1 Adopted conceptual boundary

Energy is not added to the core tuple as a sixth node coordinate. A node is not assigned an arbitrary internal battery merely to make transfer possible.

The Stage 2 environmental picture is instead an open driven system. The environment produces perturbations. Those perturbations provide excitation that can be absorbed, transmitted, used to perform structural work, or rendered less coherent through mismatch/damping.

Conceptual flow:

environmental perturbation $\rightarrow$ node excitation $\rightarrow$ interaction $\rightarrow$ transfer / recoil / work / dissipation.

5.2 Normalized bookkeeping rather than physical joules

Current Stage 2 runtimes use normalized computational units. The terms energy, work, dissipation, and entropy are bookkeeping analogies unless and until a physically dimensioned model is separately derived and validated.

This prevents two opposite errors: pretending that a simulation quantity is already real thermodynamic energy, and losing conservation/accounting discipline merely because the system is computational.

5.3 Open-system accounting

A conceptual cluster-level budget is:

$$dE_{\text{cluster}}/dt = P_{\text{env}} - P_{\text{diss}} - P_{\text{work}},$$

where environmental perturbation supplies forcing, dissipation represents excitation no longer available to coherent transfer, and work represents excitation committed to persistent structural change.

Within a discrete transfer event, the initial Stage 2 runtime enforces the bookkeeping identity:

$$E_{\text{available}} = E_{\text{dissipated}} + E_{\text{work}} + E_{\text{carried}}$$

up to floating-point tolerance.

Carried excitation becomes transient input available for later node-to-node transfer. This flow is auxiliary interaction state; it does not change the canonical node tuple.

5.4 “Costless” transfer

A perfectly aligned interaction may be modeled as approaching zero mismatch cost. That does not mean the transfer requires no excitation and does not mean energy is created.

The intended limiting interpretation is:

$$S_{\text{Theta}} \rightarrow 1 \text{ implies } C_{\text{mismatch}} \rightarrow 0,$$

and therefore E_{out} approximately equals E_{in} for the transferable portion before any work term is removed.

This is efficient transmission, not free-energy production.

6. CANDIDATE RESONANCE-TRANSFER LAW

For the first Stage 2 controlled test, the resonance-mediated arms use:

$$\begin{aligned} \eta_{ij} &= S_{\text{Theta}}(i,j) \\ C_{ij} &= 1 - S_{\text{Theta}}(i,j). \end{aligned}$$

Here η_{ij} is normalized transfer efficiency and C_{ij} is normalized mismatch cost.

These are candidate equations selected because they implement the minimum monotonic relationship suggested by the project's older harmony/low-cost idea. They deliberately avoid extra free parameters in the first test.

The candidate law does not say that resonance creates excitation. It says that resonance controls what fraction of available excitation survives an interaction as transferable output.

The frozen-efficiency control instead uses $\eta_{ij} = \text{mean } S_{\text{Theta}}(0)$, the Tick-0 mean pairwise resonance similarity, held constant for the entire control trajectory. This creates a paired comparison in which the initial efficiency scale is matched but later changes in resonance cannot alter transfer efficiency.

7. LOCAL TRANSFER SEQUENCE FOR EL-EXP-REEL-001

The first Stage 2 runtime uses the following candidate sequence:

environmental input $\rightarrow Y \rightarrow$ Theta response $\rightarrow S_{\text{Theta}} \rightarrow \eta \rightarrow Z2$ successful-transfer response / $Z1$ mismatch-recoil response $\rightarrow \Delta X$ when work occurs.

The system contains no explicit persistent preference weight for a neighbor. Any preferential pathway must arise from current node state, resonance similarity, carried excitation, and the local coupling rules.

A small amount of transfer-dependent resonance coupling is permitted: received transfer can move the destination's Theta toward the source's Theta. This is a local oscillator-style coupling rule. The experimental question is not whether that local rule exists; it is whether the closed sequence produces persistent system-level feedback differentiation relative to the control.

8. COMMISSIONING AND REPRODUCIBILITY REQUIREMENTS ADDED AFTER STAGE 1

Stage 2 laboratory work adopts the commissioning lessons established during LAB-COM-001.

The experimental apparatus should:

- boot READY / PAUSED at Tick 0;
- require an explicit Start action;
- support Pause and deterministic single-Step;
- use a visible editable seed;
- rebuild the complete initial condition reproducibly from that seed;
- include spatial state in seeded initialization when spatial state is scientifically active;
- eliminate unseeded random calls from initialization and scientific dynamics;
- prevent diagnostic/test-harness randomness from perturbing the simulation RNG;

- clear run histories at a canonical reset boundary;
- keep simulation updates independent of display frame cadence;
- ensure observation does not alter state;
- distinguish initialization controls, dynamics controls, environmental controls, observation controls, and recording controls.

Earlier commissioning work identified and corrected hidden sources of non-reproducibility, including unseeded spatial initialization and nondeterministic diagnostic sampling. These lessons become Stage 2 requirements regardless of which visual interface is used.

EL-EXP-REEL-001 adds one additional requirement: within-tick synchronous updates. Every pairwise interaction for a tick is calculated from the same frozen pre-interaction state. No node is permitted to benefit or suffer merely because its index was visited earlier in a JavaScript loop.

9. EL-EXP-REEL-001 — FIRST STAGE 2 PREREGISTERED EXPERIMENT

9.1 Motivation

The experiment is inspired by the general structure of a recommendation-feedback process:

exposure -> response/state change -> changed future interaction -> changed later exposure.

It does not attempt to reconstruct or claim knowledge of a proprietary Reel recommendation algorithm.

9.2 Research question

Does a selective environmental exposure have a different persistent effect when node-to-node transfer efficiency is dynamically resonance-dependent than when transfer efficiency is frozen?

9.3 Factorial design

All four arms begin from the exact same seeded Tick-0 node state and receive the same deterministic base environmental forcing at every tick.

Arm A — resonance-mediated + Tick-51 pulse.

Arm B — resonance-mediated + no pulse.

Arm C — frozen efficiency + Tick-51 pulse.

Arm D — frozen efficiency + no pulse.

At Tick 51 only A and C receive the additional Node-1 pulse. B and D receive base forcing only. For observable M, the pulse effect under resonance is A-B, the pulse effect under frozen efficiency is C-D, and the primary mechanism-by-pulse interaction is (A-B)-(C-D). This

difference-in-differences design prevents ordinary pre-intervention divergence between the two transfer mechanisms from being mistaken for an effect of the selective exposure.

9.4 Fixed checkpoints

0, 25, 50, 75, 100, 150.

The runtime automatically pauses at every post-zero checkpoint.

9.5 Observables

The preregistered runtime records structured serialized node state; mean X, Y, Z1, Z2 projections; pairwise Theta similarity and its variance; carried excitation; environmental input; transferred excitation; dissipated excitation; work; pathway concentration; excitation-accounting residual; pulse counts; A-B pulse effect; C-D pulse effect; and factorial interaction (A-B)-(C-D).

9.6 Result classes

A persistent post-pulse mechanism-by-pulse interaction through Tick 150 supports the narrow claim that, under these explicit model rules, selective exposure has a persistent effect that differs according to whether transfer is resonance-dependent.

A transient interaction supports mechanism-dependent perturbation sensitivity but not persistent reinforcement under these parameters.

Pulse effects with little or no factorial interaction support exposure sensitivity without a resonance-specific effect; little or no A-B or C-D response fails to support measurable pulse differentiation under the selected parameters.

A protocol violation, wrong runtime, incorrect checkpoint, nonidentical Tick-0 state, $A \neq B$ or $C \neq D$ before Tick 51, pulse counts other than 1/0/1/0 after Tick 51, or material bookkeeping error invalidates the run as evidence.

9.7 Experimental closure — EL-EXP-REEL-001

EL-EXP-REEL-001 is closed as a completed Stage 2 experiment. The declared Version B series used seeds 5, 18, 103, 449, and 871, each exported at checkpoints 0, 25, 50, 75, 100, and 150. All 5/5 B seeds passed protocol and accounting validity. An independent Version C replication executed the same canonical runtime through native Google Chrome and produced 5/5 valid C seeds. All 30 B/C checkpoint exports were byte-for-byte identical and had matching SHA-256 hashes. Version A adjudication was therefore not triggered.

Canonical reproducibility record: repository

<https://github.com/Leonardodecreative/EmergenceLab.git>; commit

3aedce78ad3c379bf5f42ecdb91dac77aea543f5; runtime stage2/reel-001/EL-EXP-REEL-001_FACTORIAL.html; runtime SHA-256 e8060e48c38d82150591cc012525f56161ee412e3961376e364173ad61092870; protocol SHA-256 272385933f8c1b406e1a0711e8317f26046b2655aecb784aeebf5072889fb48b.

9.8 Primary findings

The primary mechanism-by-pulse contrast was (A-B)-(C-D), where A-B is the effect of the Tick-51 pulse under live Theta-dependent transfer and C-D is the effect of the same pulse under frozen transfer efficiency.

Across all five seeds and all three post-pulse checkpoints (75, 100, 150), cumulative modeled transfer interactions were positive, cumulative modeled work interactions were positive, and cumulative modeled dissipation interactions were negative. At Tick 150 the cross-seed mean interactions were +12.3575 transferred, +1.0442 work, and -1.0142 dissipated in normalized computational units. These are computational bookkeeping quantities, not physical joules.

The strongest replicated result is therefore not monotonic Theta convergence. Mean Theta similarity was non-monotonic in 13 of 20 seed-by-arm trajectories, and the Tick-150 Theta interaction was mixed across seeds. Pathway-concentration effects were very small, while retained-flow and last-tick effects were sign-variable. The data support the narrower conclusion that, in this runtime, allowing transfer efficiency to respond to current relational Theta state reproducibly changes how the same perturbation is processed, with more cumulative transfer and work and less cumulative dissipation than the frozen-efficiency control.

9.9 Identity and architectural interpretation

Final X vectors remained numerically distinct: there were zero exact pairwise X-vector collisions across the 20 seed-by-arm final states. However, only 70 of 80 final node X vectors remained closest to their own Tick-0 X vector under the self-nearest operational test. Strong identity-signature preservation is therefore not universal and requires dedicated follow-up if it becomes a primary claim.

The result is compatible with Theta functioning as relational compatibility/coupling rather than as a rule whose purpose is to make nodes identical. That interpretation is supported but not uniquely proven. REEL-001 does not establish literal thermodynamics, metabolism, canonical meanings for Z1 or Z2, reward learning, dopamine-like motivation, prediction, recursive learning, consciousness, or a proprietary recommendation algorithm.

REEL-001 belongs to the relational node-to-node interaction/self-definition hypothesis family. It does not test the separate reward/dopamine-like feedback family or the separate prediction family.

9.10 Relation to the recursive closed-loop program

REEL-001 does not validate the full Reality-to-Action recursive loop. It supplies evidence for a narrower internal transition mechanism: environmental input can alter retained node state, current Theta compatibility can alter transfer efficiency, Z1/Z2-mediated processing can partition the perturbation differently into transfer, dissipation, and work, and modeled work can contribute to X change. In the closed-loop program this should be treated as evidence about a submechanism connecting input/memory-bearing state toward identity update, not as evidence that consequential weighting, meaning, purpose, prediction, reward, or action selection have already been demonstrated.

10. EXPLICIT EXCLUSION OF RESTORED-ANATOMY V2 FROM THIS TEST

Stage 2 experimental minimalism must not be confused with rejection of the project's restored anatomical vision.

The separate V2 anatomy lineage includes concepts such as Mati / central core; radials and axes as rendered anatomy; orbital/circulatory movement; resonance rings; richer three-dimensional spatial expression; and visual transfer pathways.

Those features may later become scientifically causal. EL-EXP-REEL-001 excludes them because the immediate question can be asked with a smaller causal surface. If a later experiment tests anatomical mechanisms, they must be introduced explicitly as independent structures rather than silently inherited by this runtime.

Therefore: Stage 2 white paper is not the restored-anatomy V2 brief. The documents address different project layers and remain separate until a later deliberate integration decision.

11. GAMMA AND CONSEQUENTIAL HISTORY

The project has also developed the concept of Gamma as consequentiality, legacy, scarring, or forward influence from prior events.

Stage 2 preserves Gamma as a relevant research concept but does not make it causal in EL-EXP-REEL-001. The first resonance-transfer experiment should determine what the five-coordinate node plus environmental forcing can do without adding a dedicated long-history reinforcement variable.

This is an intentional control on architectural complexity. If persistent feedback cannot survive without a Gamma-like term, that itself becomes useful evidence for a later experiment.

12. WHAT STAGE 2 DOES NOT YET CLAIM

Stage 2 does not establish that the model is physically thermodynamic; its normalized excitation is literal energy; its environmental disorder parameter is thermodynamic entropy; the selected

transfer law is uniquely correct; Z1 is identical to mechanical inertia; Z2 is identical to kinetic energy; resonance alone produces learning; observed persistence constitutes consciousness or intelligence; the model reproduces any proprietary recommendation algorithm; or restored-anatomy structures are merely decorative or scientifically unnecessary.

These boundaries preserve falsifiability and keep the project from confusing a useful analogy with a demonstrated equivalence.

13. UNRESOLVED STAGE 2 QUESTIONS

1. What functional form should replace or refine $\eta = S_{\text{Theta}}$ if the linear mapping is inadequate?
2. Should transfer cost be modeled solely as mismatch loss, or should there be an additional activation threshold?
3. What is the best operational distinction between recoil, recovery, damping, and structural repair inside Z1?
4. What is the best operational distinction between transfer acceptance, integration, and carryover inside Z2?
5. Under what conditions should work alter X, and should that work be reversible?
6. How should incoherent/dissipated computational excitation re-enter the environment, if at all?
7. What conservation law, if any, should be imposed on a later physically dimensioned model?
8. Does Gamma add explanatory power beyond path dependence already present in X, Y, Z1, Z2, Theta?
9. Can the high-disorder driven system self-organize into resonant clusters without an explicit clustering objective?
10. Does the restored anatomy change measurable dynamics once introduced causally, or does it remain representational?
11. How sensitive are Stage 2 outcomes to vector dimensionality, seed, population size, and intervention target?
12. Can the system sustain useful circulation of excitation while still paying work/dissipation costs without pathological numerical collapse or growth?

14. STAGE 2 RESEARCH DOCTRINE

Stage 2 adopts the laboratory rule:

Freeze the specimen, define the intervention, preregister the checkpoints, run the control, and interpret only after the run.

New architecture is not to be introduced mid-experiment because an observation is interesting. Interesting observations are recorded and parked for later deliberation.

The Stage 2 objective is not to make the node look intelligent. It is to determine what organization, persistence, recovery, differentiation, and feedback can arise from a minimal explicit substrate under controlled perturbation.

15. CURRENT STAGE 2 IMPLEMENTATION RECORD

A dedicated branch has been created:
stage2/reel-001

inside:
leonardodecreative/EmergenceLab

The canonical experiment runtime is:
stage2/reel-001/EL-EXP-REEL-001_FACTORIAL.html

The final runtime starts READY / PAUSED at Tick 0; uses deterministic seeded initialization; creates four identical Tick-0 arms; derives base environmental forcing from seed + tick; applies the selective pulse only to A and C at Tick 51; uses structured X, Y, Z1, Z2 and Theta state; contains no Mati/radial/orbital dynamics; treats excitation as auxiliary flow rather than a node coordinate; records environmental input, transfer, dissipation, work and carry; calculates pair interactions synchronously from frozen pre-interaction snapshots; pauses automatically at preregistered checkpoints; displays integrity checks and factorial contrasts; and exports complete JSON snapshots and telemetry.

The earlier index.html and EL-EXP-REEL-001.html files are retained as development audit artifacts. They are superseded and must not be used to generate evidence. The canonical runtime is the explicitly named FACTORIAL file above.

16. CURRENT STATUS AND NEXT EXPERIMENTAL SEQUENCE

EL-EXP-REEL-001 is complete and independently replicated. No Version A adjudication was required.

The immediate programmatic priority is to complete the two remaining previously separated hypothesis families before expanding either into deeper sub-hypotheses: reward/dopamine-like feedback and prediction. Each must receive its own minimal preregistered design and independent B/C replication rather than being folded into REEL-001 retroactively.

REEL-001 remains the closed evidence base for the relational node-to-node interaction/self-definition family.

Only after the three original hypothesis families are individually closed should the laboratory pursue deeper branches within the prediction hypothesis or other rabbit-hole extensions.

END OF STAGE 2 WHITE PAPER — WORKING REVISION 3

Stage 1 remains frozen. Restored-anatomy V2 remains separate. Candidate Stage 2 mechanisms remain subject to experiment.

ADDENDUM A

GAMMA CANDIDATE EXPERIMENTAL ARCHITECTURE

Consequential-history storage, structured causal readout, relational indexing, and falsification boundary

Status	Candidate instrument and experimental architecture; not canonical node anatomy
Date	10 September 2026
Evidence boundary	Supplements Section 11. Does not revise the closed results of EL-EXP-REEL-001.
Implementation boundary	Disposable Gamma draft only. The canonical engine remains unchanged.

A.1 PURPOSE AND CORRECTION OF THE FALSIFICATION BOUNDARY

Stage 2 previously preserved Gamma as an unresolved consequential-history concept while deliberately excluding it from EL-EXP-REEL-001. The present addendum specifies an apparatus for testing that concept without promoting Gamma into the canonical node or assigning it a causal role in advance.

A negative result from one Gamma experiment does not eliminate an architecture class. It rejects only the tested combination of measurement instrument, state representation, retrieval rule, placement, parameterization, and experimental conditions. Neighboring Gamma architectures remain open unless separately tested.

The laboratory must also distinguish a failure to measure consequence, a failure to represent consequence, a failure to retrieve relevant history, and a genuine failure of Gamma to add explanatory or causal value. These are not interchangeable outcomes.

A.2 CONSTRUCT SEPARATION

Earlier discussion placed causality, influence, meaning, valuation, legacy, and scarring near Gamma. Stage 2 now separates them. The candidate instrument operationalizes causal consequence as divergence between matched future trajectories with and without a specified event. It does not operationalize moral value or subjective meaning.

Construct	Stage 2 treatment
Causal reach	Measured factual-versus-counterfactual trajectory difference
Influence	Observed propagation of an event across state, relation, or environment
Meaning	Unresolved; not inferred merely from high causal magnitude
Valuation	Separate normative or selection problem; not stored as causal truth
Legacy / scarring	Persistent consequences that may be represented within settled causal history

A.3 THE COMPUTER, THE ARCHIVE, AND THE READOUT

Gamma is not assigned the simultaneous roles of processor, storage medium, and experimental record. The candidate architecture separates three operators from one stored state:

```
C(event) -> c_vector
Gamma_i(t+1) = B_K(Gamma_i(t), settled_record)
G_vector_i(t) = Q(Gamma_i(t), present_pattern, relational_query)
```

- C is the external causal-measurement instrument. It clones a matched state and noise stream, permits the event in one branch, suppresses only that event in the other, and measures downstream divergence over horizon H.
- Gamma is a bounded archive of settled causal products. It is not a complete episodic memory and does not retain every intermediate operation.
- B_K is the deterministic archive update and capacity rule.
- Q is the context-sensitive read operator that performs pattern matching and produces simultaneous channel and counterpart readouts.
- L, the external experimental ledger, stores complete provenance: event data, branch states, suppressions, trajectory exports, measurements, and validation results.

A.4 PROVISIONAL NON-SCALAR GAMMA STATE

The current candidate state is a bounded, variable-membership archive with a fixed and versioned record schema:

```
Gamma_i(t) = { (s_ik, c_vector_ik, tau_ik, H_ik, q_ik, rho_ik) } for k = 1...K
```

Field	Function
s	Pattern signature at the event or settlement context
c_vector	Signed normalized factual-minus-counterfactual consequence vector
tau	Settlement timestamp or tick
H	Measurement horizon
q	Reliability or confidence attached to the measurement
rho	Relational metadata: actor, recipient, counterparts, interaction, and perspective

The disposable implementation uses 23 signed consequence channels spanning the implemented X, Y, Z1, Z2, Theta, relational-influence, and environmental families. This dimensionality is an experimental schema, not a discovered natural constant. Gamma's archive membership and stored values may change continuously while the schema remains fixed within an experiment. This is open content inside a closed ruler.

A.5 STRUCTURED RETRIEVAL

For kernel match k_{ik} between the present pattern $p_i(t)$ and stored signature s_{ik} , the provisional readout is:

```
R_i(t) = min(1, sum_k q_ik k_ik)
G_vector_i(t) = R_i(t) [sum_k q_ik k_ik c_vector_ik] / [epsilon + sum_k q_ik k_ik]
```

G_vector remains structured. A scalar norm may be displayed as telemetry, but it is not substituted for Gamma or used as proof of meaning. All channels and per-counterpart read heads are computed from the same frozen pre-commit state and promoted together on the next cycle. This is logical synchronous multi-computation; it is not a claim of hardware-level parallel execution.

A.6 RELATION TO Y, X, WRITE SUCCESS, AND NODE INTERACTION

The candidate Gamma architecture does not replace Y or X. Y remains the memory/perception and retained-interaction pathway. X remains comparatively persistent identity organization. A qualified Y-to-X attempt may succeed or fail through the Z2/Z1 pathways. Gamma settlement occurs later and depends on measured downstream consequence, not on the mere occurrence or success of a write attempt.

A -> B input -> Y_B -> qualification / transfer attempt -> [X_B integration or Z1/Z2 response] -> measured downstream trajectory -> optional settled Gamma record

Therefore, a failed Y-to-X write may still earn a Gamma record if it measurably changes later memory, recoil, integration pressure, resonance, relationship state, or environmental trajectory. Conversely, a successful write does not automatically earn a high Gamma record if its later causal reach is negligible. No write count, coherence value, X value, or failure flag is accepted as a substitute for the counterfactual consequence measurement.

A.7 STORAGE OWNERSHIP AND PAIRWISE HISTORY

Information	Provisional location
Experienced input and temporal trace	Y of the receiving or observing node
Successfully integrated persistent organization	X
Settled consequence affecting a node	That node's bounded Gamma archive
A-B versus A-C/A-D distinction	Relational metadata and counterpart-specific Gamma readouts
Complete computational and counterfactual history	External experimental ledger L
Central energetic origin / core	Mati or Quanta nucleus; explicitly not reassigned as storage

The present instrument can distinguish tagged A-B, A-C, and A-D consequential records without creating permanent edge-state objects. Whether durable pair-specific history requires a separate relational state Gamma_{ij} is an unresolved experimental fork. The node-local tagged model and an explicit edge-archive model must be compared before either is adopted.

Implementation limitation: the mounted exploratory engine presently represents X and Y numerically even though the Stage 2 conceptual architecture treats them as structured coordinates. It therefore does not yet store full semantic interaction objects inside Y. Gamma must not be used to conceal or counterfeit that missing representation.

A.8 CURRENT IMPLEMENTATION STATUS

- A disposable Gamma vector draft exists; the canonical engine remains unchanged.
- Gamma is telemetry plus externally settled history only. It does not alter X, Y, Z1, Z2, Theta, coupling strengths, movement, or the identity-write predicate.
- With Gamma disabled, and with Gamma enabled but empty, 200-step live-engine trajectories matched the canonical engine exactly under the tested seed.
- The integration harness passed 21 of 21 checks, including signed-vector retrieval, counterpart separation, cross-cycle promotion, deterministic capacity, reset behavior, validation, ablation, and absence of spontaneous records.

- The factual/counterfactual causal-measurement runner is not yet implemented. Accordingly, the settlement interface can accept externally measured consequences, but no current runtime result constitutes evidence that Gamma has earned a causal role.

A.9 REQUIRED EXPERIMENTAL SEQUENCE

Phase	Question	Failure means
0 - Calibration	Can the instrument recover deliberately injected known consequences?	Measurement failure; no claim about Gamma architecture
1 - Representation	Does structured Gamma retain predictive information beyond X, Y, Z1, Z2, Theta, recency memory, and hysteresis?	This representation or feature schema is inadequate or redundant under the tested conditions
2 - Retrieval	Does context-sensitive retrieval activate the correct prior consequences?	The tested kernel, relational query, or readout law is inadequate
3 - Causal role	Does enabling a preregistered Gamma pathway reproducibly change later trajectories beyond controls?	This placement or causal interface lacks supported effect under the tested conditions
4 - Robustness	Does the result survive seeds, parameter neighborhoods, population changes, ablation, and independent B/C replication?	The result is brittle, local, or unreplicated

A.10 ACCEPTANCE AND NON-ACCEPTANCE RULES

Gamma does not earn a place because the archive can be populated, because a readout is nonzero, or because an intervention produces an interesting trajectory. Promotion requires a valid instrument, prespecified comparison, unique out-of-sample predictive or causal contribution beyond simpler alternatives, robustness across matched runs, and independent B/C replication. Version A remains reserved for adjudicating B/C disagreement.

The relevant predictive comparison is whether the future is better distinguished when Gamma is included than when the implemented node and simpler history mechanisms are used alone:

$P(\text{future} \mid X, Y, Z1, Z2, \text{Theta}, \text{Gamma})$ versus $P(\text{future} \mid X, Y, Z1, Z2, \text{Theta}, \text{simpler-history controls})$

No single failed implementation authorizes the statement that consequential-history architectures as a whole have been eliminated. The strongest permitted negative conclusion is local: the tested Gamma realization did not add supported value under the specified protocol and parameter region.

A.11 WORKING-DOCUMENT STATUS CORRECTION

Where Working Revision 3 states that Stage 1 itself remains frozen, the intended distinction is hereby clarified: executed Stage 1 runtimes, seeds, protocols, and recorded results remain frozen for reproducibility; the Stage 1 white paper remains open to review and correction. This addendum does not alter any recorded experimental result.

END OF ADDENDUM A - STAGE 2 WORKING REVISION 4

ADDENDUM B**EL-EXP-GAMMA-001 v1.5 - 30-SEED EVALUATION**

Predictive value of structured consequential history under a frozen noncausal protocol

Status	Expanded evaluation completed; statistical signal supported; causal promotion not authorized
Execution date	11 September 2026
Evidence boundary	Gamma remained a noncausal predictor. Canonical X, Y, Z1, Z2, Theta dynamics were not changed.
Frozen build	EL-EXP-GAMMA-001-v1.5.0
Core SHA-256	33d4c44561382c581625aa801347b5da7d7e1c901b2cb6cdf9694b1211ead85
Config hash	13094d04
Scientific result SHA-256	c8ebcc011a057fd0f00d0a89379cac4843fb61fe1c1f98d480826ee7a3182c16

B.1 PURPOSE AND DECISION RULE

This experiment tested whether the candidate Gamma archive and relational retrieval rule retain out-of-sample information about measured downstream consequence beyond the implemented node state and simpler history controls. It did not test whether Gamma should directly alter node dynamics.

The primary null hypothesis was that, across seeds, positive and negative Gamma improvement relative to the best ordinary control are equally probable. Separately, practical advancement required both frozen effect margins: at least 5% pooled balanced-RMSE improvement over the best ordinary control and at least 10% over shuffled Gamma history.

B.2 FROZEN FIELD AND PROCEDURE

Element	Frozen value or procedure
Seeds	1-30 in three batches: 1-10, 11-20, 21-30
Nodes	6
Warmup	30 ticks
Acquisition / test	360 / 180 events per seed
Consequences	Signed 23-channel factual-minus-counterfactual vector
Horizon	10
Archive capacity	48
Kernel bandwidth	0.015
Strength levels	0.12, 0.2, 0.28
Ordinary controls	zero, recency, typeMean, pairTypeMean, stateKernel
Gamma models	gammaDyadic and gammaRelational
Permutation control	shuffledGammaDyadic and shuffledGammaRelational
Replication order	Bobby B first; Codex C second; 12-decimal rounded scientific hashing

Each seed contributed 180 held-out test events; pooled model metrics therefore cover 5,400 predictions. The best ordinary comparator was selected by lowest pooled balanced RMSE from the prespecified ordinary-control set.

Protocol qualification

- The three batch files matched the frozen build identifier, configuration hash, and core SHA-256.
- Seeds were unique and complete from 1 through 30.
- Seeds 5 and 18 overlap the earlier five-seed pilot. The N=30 result is therefore an expanded evaluation, not a wholly independent confirmatory sample.
- Exact sign tests discarded no ties. Confidence intervals for mean improvement use a two-sided t interval across seeds; positive-rate intervals use Wilson's method.

B.3 AGGREGATED RESULTS

Comparison	Seed signs	Mean improvement (95% CI)	Median	Exact two-sided p
Gamma vs best ordinary	22 + / 8 - / 0 ties	2.34% (0.90% to 3.78%)	2.01%	0.016125
Gamma vs shuffled	29 + / 1 - / 0 ties	8.60% (6.99% to 10.21%)	8.19%	5.774e-08

Pooled measure	Observed	Frozen requirement	Decision
Best ordinary control	typeMean; RMSE 0.006764	Prespecified control set	Reference
Gamma relational RMSE	0.006604	Lower is better	Best pooled RMSE
Improvement vs best ordinary	2.37%	>= 5.00%	FAIL margin
Shuffled Gamma RMSE	0.007229	Permutation control	Reference
Improvement vs shuffled	8.65%	>= 10.00%	FAIL margin
Seeds passing both margins	7 / 30	Pilot rule was 4 / 5	Descriptive only

The exact two-sided sign test rejects the primary null at alpha = 0.05. Gamma v1.5 has a reproducible positive predictive direction across seeds. However, it fails both frozen practical effect margins. The correct verdict is therefore: signal detected; causal promotion withheld.

Statistical significance and laboratory acceptance answer different questions. The sign test asks whether the direction is unlikely under a 50/50 null. The frozen margins ask whether the effect is large enough to justify advancing this implementation. The first answer is yes; the second is no.

Positive-rate intervals

- Versus best ordinary control: 73.33%; 95% Wilson interval 55.55% to 85.82%.
- Versus shuffled history: 96.67%; 95% Wilson interval 83.33% to 99.41%.

B.4 SEED-LEVEL RESULTS

Improvement is (control RMSE - Gamma RMSE) / control RMSE. Positive values favor Gamma. 'Pass' is the frozen per-seed conjunction used by the v1.5 runner; it is reported descriptively and does not replace the pooled decision.

Seed	Best control	vs control	vs shuffled	Pass
1	typeMean	-1.31%	8.01%	no
2	typeMean	2.33%	5.45%	no
3	typeMean	8.66%	14.31%	yes
4	typeMean	0.18%	11.34%	no
5	typeMean	-7.35%	0.82%	no
6	typeMean	-0.55%	7.40%	no
7	typeMean	2.42%	5.71%	no
8	typeMean	-0.23%	6.30%	no
9	typeMean	0.80%	10.51%	no
10	typeMean	4.76%	12.90%	no
11	typeMean	6.79%	11.08%	yes
12	typeMean	1.07%	6.17%	no
13	typeMean	-1.96%	2.31%	no
14	typeMean	-2.22%	6.69%	no
15	typeMean	7.27%	12.02%	yes

Seed	Best control	vs control	vs shuffled	Pass
16	typeMean	-2.84%	-0.11%	no
17	typeMean	1.09%	8.37%	no
18	typeMean	5.42%	10.67%	yes
19	typeMean	3.30%	14.04%	no
20	typeMean	7.01%	12.06%	yes
21	typeMean	0.71%	4.72%	no
22	typeMean	4.59%	7.51%	no
23	typeMean	7.31%	16.09%	yes
24	typeMean	4.30%	9.56%	no
25	typeMean	1.70%	12.73%	no
26	typeMean	9.74%	16.99%	yes
27	typeMean	0.67%	6.98%	no
28	typeMean	4.26%	5.82%	no
29	typeMean	-1.12%	2.99%	no
30	typeMean	3.29%	8.68%	no

Range and heterogeneity

Improvement over the best ordinary control ranged from -7.35% to 9.74% (sample SD 3.86%); improvement over shuffled history ranged from -0.11% to 16.99% (sample SD 4.31%). The negative seeds and wide range show that the added information is heterogeneous and not uniformly beneficial under the tested field.

B.5 WHAT THE RESULT SUPPORTS

- A bounded relational consequential-history archive contains predictive information not fully captured by the tested ordinary controls under this field.
- Correct signature-consequence pairing matters. The 29-of-30 advantage over shuffled history is substantially stronger than the advantage over the best ordinary control.
- Event type alone is already powerful: typeMean was the best ordinary baseline and nearly matched Gamma. Gamma's additional context signal is real but modest.
- Gamma is better treated as a structured archive plus a contextual read operator than as a single scalar state. The experiment does not show that all meaning, causality, influence, valuation, or legacy are one variable.
- The result preserves the Gamma architecture class. It evaluates only v1.5's schema, bandwidth, horizon, capacity, retrieval rule, and field.

B.6 WHAT THE RESULT DOES NOT SUPPORT

- It does not authorize Gamma to modify X, Y, Z1, Z2, Theta, coupling strengths, or the identity-write gate.
- It does not establish subjective meaning, moral valuation, consciousness, or metaphysical causality.
- It does not prove that Gamma is a missing universal component of every node architecture.
- It does not establish a wholly independent confirmation because two seeds overlapped the pilot and the analysis followed prior exploratory development.
- It does not determine whether the missed margins arise from weak construct value, lossy representation, poor retrieval geometry, inappropriate timescale, limited capacity, or field mismatch.

B.7 PRIMARY INFERENCE

Gamma v1.5 has earned continued experimentation, not membership in the canonical node. The laboratory has moved from 'does consequential history contain any recoverable signal?' to 'where is that signal located, why is it lost on some seeds, and can one prespecified repair strengthen it without overfitting?'

B.8 NEXT EXPERIMENTAL STEP

The next step is a noncausal diagnostic decomposition using the frozen N=30 outputs. No equations should be changed during this analysis. Treat all subgroup findings as exploratory and use them only to nominate one v1.6 repair.

Diagnostic	Question
Event class	Is gain concentrated in particular intervention types or strength levels?
Consequence family	Which of X, Y, Z1, Z2, Theta, relational, or environment channels carry the signal?
Actor-target relation	Does relational Gamma help particular dyads or interaction directions?
Gamma form	Where does gammaRelational outperform or lose to gammaDyadic?
Archive dynamics	Do capacity saturation, merge, eviction, or record age predict failures?
State regime	Are gains associated with coherence, recovery, perturbation, or boundary regions?

After that decomposition, preregister one specific v1.6 candidate and test it on 30 fresh, hash-derived seeds excluding 1-30 and the pilot-only seeds 103, 449, and 871. Preserve Bobby B first and Codex C second, frozen code/config hashes, exact sign tests, interval estimation, shuffled controls, and practical advancement margins.

B.9 NEW QUESTIONS OPENED BY THE RESULT

- Does one global kernel bandwidth ($\sigma = 0.015$) underweight some state families and overfit others?
- Does the 48-record capacity discard older but highly consequential history, or does a larger archive merely add noise?
- Does the single horizon $H = 10$ hide short- and long-timescale consequences?
- Should consequences be retrieved through separate family-specific or counterpart-specific heads rather than one aggregate relational kernel?
- Is typeMean absorbing genuine causal structure because event class dominates magnitude? Would residual prediction beyond typeMean be a cleaner target?
- Which signed consequence channels account for the advantage, and do they replicate after correction for multiple comparisons?
- Why do eight seeds favor the ordinary control, and what pre-event regime distinguishes them?
- Will the direction and magnitude survive a fully fresh seed set and an independently rebuilt runner?
- If a later noncausal candidate clears the margins, does a preregistered causal Gamma pathway improve recovery, adaptation, or counterfactual accuracy without destabilizing the node?

B.10 REPRODUCIBILITY AND PUBLIC CODE

The frozen v1.5 code, protocols, runners, batch outputs, aggregator, final summary, and checksums should be deposited in the Emergence Lab repository as a versioned historical experiment. They should not be merged into the canonical engine or described as production Gamma.

Repository item	Required content
README	Question, architecture boundary, commands, decision rules, verdict, limitations
src	Frozen v1.5 core identified by SHA-256
protocol	Protocol and calibration documents
runners	Batch scripts and final aggregation script
results	Three batch JSONs and final summary JSON
checksums	Manifest covering every public artifact
release metadata	Runtime versions, replication order, seed overlap, noncanonical status

Recommended repository path: experiments/gamma/EL-EXP-GAMMA-001-v1.5/. Tag the release after verifying that a clean checkout reproduces the rounded scientific SHA-256. Preserve the current v1.6 exploratory source separately; do not relabel it as v1.5.

B.11 STAGE 2 VERDICT

EL-EXP-GAMMA-001 v1.5 rejects the equal-sign null and demonstrates a small, repeatable predictive contribution from correctly indexed consequential history. It fails the laboratory's prespecified practical thresholds. Gamma therefore earns a second experimental round and a diagnostic map; it does not yet earn a causal socket in the canonical node.

END OF ADDENDUM B - STAGE 2 WORKING REVISION 5

ADDENDUM C

GAMMA v1.5 DIAGNOSTIC AND v1.6 PREREGISTRATION

Localization of predictive signal and nomination of one retrieval-geometry repair

C.1 STATUS AND EVIDENCE BOUNDARY

The frozen N=30 decision recorded in Addendum B remains unchanged. The present analysis is exploratory, noncausal, and restricted to locating the predictive signal already present in the sealed v1.5 output. It does not revise the acceptance margins or install Gamma in the canonical node.

- All 30 final-world digests matched the sealed batch outputs.
- Pooled Gamma and typeMean RMSE matched the sealed aggregation.
- The analysis covered 5,400 held-out test events from the frozen 360-acquisition / 180-test schedule.
- Benjamini-Hochberg q-values were calculated separately within diagnostic families. Subgroup results remain exploratory.

Diaagnostic scientific SHA-256

bffcee877c6c91aca08f526e8856ec3b14c6cb4c67458b0298abe72843b848ab

C.2 PRIMARY DECOMPOSITION

Comparison or stratum	Pooled improvement	Seed direction	Interpretation
Gamma vs typeMean	2.37%	22 / 30 positive	Reproduces frozen primary result
Gamma vs shuffled history	8.65%	29 / 30 positive	Correct signature-consequence pairing matters
Relational vs dyadic Gamma	31.85%	30 / 30 positive	Context across relations is load-bearing
Theta consequence family	8.72%	29 / 30 positive	Most reliable consequence-family gain
Relation consequence family	4.71%	26 / 30 positive	Secondary structured gain
DISRUPT_PHASE	8.08%	29 / 30 positive	Strongest event-class gain
ALIGN_THETA	-1.76%	7 / 30 positive	Systematic retrieval weakness

The decomposition does not show a general Gamma advantage across every channel. Z2, environment, amplitude, and several relational channels were invariant or carried no measurable improvement in this field. The useful signal is concentrated rather than universal.

C.3 PHASE-GEOMETRY DEFECT

The v1.5 signature kernel computes ordinary squared distance for every coordinate. Normalized phase is not linear: 0 and 1 represent the same angular boundary. Consequently, phases 0.99 and 0.01 are physically separated by 0.02 but appear 0.98 apart to the v1.5 metric.

```
delta_circ(a,b) = min( |a-b|, 1-|a-b| )
```

The diagnostic pattern is consistent with this defect: phase-disruption history is highly informative, yet alignment events degrade relative to typeMean. The result does not prove that circular distance will repair the deficit. It provides a principled, falsifiable nomination that does not require coefficient fishing.

C.4 SOLE v1.6 CHANGE

EL-EXP-GAMMA-002 v1.6 changes only the component distance used for six normalized phase coordinates during Gamma signature retrieval:

Signature section	Circular indices	Fields
Target context	10, 11, 12	Theta.phase.0, Theta.phase.1, Theta.phase.2
Actor state	33, 34, 35	Theta.phase.0, Theta.phase.1, Theta.phase.2

- All non-phase coordinates retain linear distance.
- Sigma remains 0.015; capacity remains 48; horizon remains 10.
- The event schedule, measured consequences, state dynamics, archive writes, and final-world digests must remain identical to v1.5.
- Only Gamma predictions and derived metrics may change.
- No causal pathway into X, Y, Z1, Z2, Theta, coupling, or the write gate is permitted.

C.5 FRESH CONFIRMATORY FIELD

Thirty fresh seeds were derived and fixed before implementation using the first four big-endian bytes of SHA-256 over the protocol identifier and positive integer index. Seeds 1-30 and pilot-only seeds 103, 449, and 871 were excluded. Bobby B runs first; Codex C runs second; Version A adjudicates a material discrepancy.

No seed may be replaced because of result direction or apparent outlier behavior. Commissioning uses excluded Seed 13579 and is non-evidentiary.

C.6 v1.6 ADVANCEMENT GATE

Circular-phase Gamma advances only if all conditions pass:

Gate	Requirement
Ordinary-control magnitude	$\geq 5.00\%$ pooled balanced-RMSE improvement over the best prespecified ordinary control
Permutation magnitude	$\geq 10.00\%$ pooled improvement over shuffled relational Gamma
Repair-specific direction	Exact two-sided seedwise sign test vs matched linear-phase Gamma at $\alpha = 0.05$, positive direction
Primary direction	Exact two-sided seedwise sign test vs best ordinary control at $\alpha = 0.05$, positive direction
Independent replication	Bobby B and Codex C agree after 12-decimal rounding on scientific payload and final hash

Failure of any gate yields DO_NOT_ADVANCE_CIRCULAR_PHASE_GAMMA_V1_6. Secondary Theta, phase, event-class, and strength analyses cannot rescue a failed primary decision.

C.7 INTERPRETATION

- A pass would support circular phase geometry as a better noncausal retrieval realization.
- A pass would authorize drafting a separate causal experiment, not modifying the canonical engine.
- A failure would reject this repair under the frozen field, not consequential-history architectures as a class.
- Neither outcome establishes subjective meaning, moral valuation, consciousness, or metaphysical causality.

C.8 REPRODUCIBILITY STATUS

The frozen v1.5 source, protocols, batched $N=30$ runners, results, aggregation, diagnostic code, decomposition, checksums, and v1.6 preregistration are preserved on a separate GitHub branch and draft pull request. The public record modifies no canonical engine file. The complete Stage 2 working paper remains in the controlled project record.

v1.6 preregistration SHA-256

```
b8ea9d564fc135e782cf65b2ffea252e9bb0a3854fab4ba3f7e4d72c2505e55a
```

C.9 STAGE 2 DECISION

Gamma v1.5 has earned a targeted second experiment. The laboratory will test one geometric repair against the unchanged linear v1.5 comparator on a fresh field. Until that protocol passes every frozen gate, Gamma remains a structured consequential-history hypothesis rather than canonical node anatomy.

END OF ADDENDUM C - STAGE 2 WORKING REVISION 6

ADDENDUM D

GAMMA v1.6 CONFIRMATORY RESULT

Thirty-seed circular-phase retrieval test, independent replication, and Version A adjudication

D.1 EXECUTION STATUS

EL-EXP-GAMMA-002 v1.6 was frozen before confirmatory execution. Bobby B executed the preregistered 30-seed field first. Codex C independently executed the same source, configuration, seed order, and analysis second. Version A adjudicated the only material record discrepancy. No canonical engine file was modified.

- JavaScript commissioning: 15 / 15 PASS; statistical tests: 3 / 3 PASS.
- Thirty unique preregistered seeds were executed in three deterministic batches per replication.
- Core, protocol, configuration, build, seed order, pooled comparisons, gates, and verdict agreed between B and C.

D.2 FROZEN PRIMARY RESULT

Comparator	Pooled RMSE	Improvement	Seed signs	Exact two-sided p
Best ordinary: typeMean	0.006513370444	0.877757%	17+ / 13- / 0=	0.584664712
Linear-phase Gamma	0.006456199466	0.000009%	7+ / 4- / 19=	0.548828125
Shuffled-history Gamma	0.007013929353	7.951755%	30+ / 0- / 0=	1.86265e-9
Circular-phase Gamma	0.006456198874	reference	-	-

The frozen advancement gate required all five conditions to pass. Four substantive conditions failed: ordinary-control magnitude, permutation magnitude, repair-specific direction, and primary direction. The adjudicated independent-replication gate passed. Final verdict:

DO_NOT_ADVANCE_CIRCULAR_PHASE_GAMMA_V1_6

D.3 VERSION A ADJUDICATION

The original Bobby B and Codex C scientific hashes differed. Inspection localized the discrepancy to three raw final-world fingerprints and 58 floating-point values whose maximum absolute difference was 1.1102230246251565e-15. Every numeric difference disappeared under the preregistered 12-decimal cross-runtime rounding rule.

- The v1.6 final-world fingerprint hashed raw JavaScript floating-point state instead of the rounded canonical state.
- Removing only those raw fingerprints and their derivative batch hashes, then applying 12-decimal rounding, made the complete role-neutral scientific payloads byte-identical.
- Adjudicated scientific SHA-256 for both replications:
9ed3471b9d203e4494cde80f5b41c0c7632e03c56bcb43383ff71e0159e2f037.

Version A classifies this as a fingerprint-instrumentation defect, not a scientific replication failure. Future cross-runtime world digests must hash a canonical 12-decimal-rounded world representation. The frozen v1.6 files and raw mismatch record remain unchanged.

D.4 WHAT THE FINDING ESTABLISHES

- Circular treatment of normalized phase is mathematically appropriate at the wrap boundary, but it was empirically inert in this field: its pooled gain over linear phase was approximately 0.000009%.
- Correct relational history retained predictive information: circular Gamma beat shuffled-history Gamma on all 30 seeds.
- That relational signal did not clear the frozen magnitude requirement against the best ordinary control.
- The current nearest-neighbor consequential-history realization is therefore not ready for causal installation in the node.
- The result rejects this circular-phase repair under the frozen field. It does not eliminate Gamma, consequential history, non-scalar archives, or the broader architecture class.

D.5 WHAT THE FINDING DOES NOT ESTABLISH

- It does not show that causality, influence, meaning, value, legacy, and scarring are one variable.
- It does not show that those functions must be separate variables.
- It does not test prospective Gamma feedback into qualification, identity writes, or node-to-node dynamics.
- It does not authorize coefficient tuning, seed replacement, or retroactive changes to the v1.6 field.

D.6 NEXT SCIENTIFIC QUESTION

The next move is diagnostic, not another immediate repair. The laboratory should measure when the v1.6 retrieval mechanism had enough usable archive support to differ from ordinary controls: kernel mass, effective matching-entry count, wrap-boundary activation frequency, event-family coverage, and the source of the 19 linear-versus-circular ties. This analysis must use the completed record and must not alter the frozen verdict.

Only after that localization should a new architecture be nominated. Candidate directions include multiscale retrieval, typed consequence channels, actor-target interaction histories, or a separate settlement-and-storage process. None is canonical until preregistered and tested. A causal Gamma experiment remains deferred because v1.6 did not clear its advancement gate.

D.7 REPRODUCIBILITY RECORD

The GitHub evidence branch contains the frozen source, preregistration, commissioning suite, deterministic runners, Codex C batches, Bobby B final summary, raw B/C comparison, and Version A ruling. The final evidence package is preserved separately from the runnable Bobby B package.

Bobbv B reported scientific SHA-256

```
8d3b01c54ef102c824f45218093a2e0caf1f38b3f918211a05093259195d974b
```

Codex C reported scientific SHA-256

```
9cd8a1413510f62be17a111a63282cd6159663ee05d25e9b6c54b6b804b972c0
```

D.8 STAGE 2 DECISION

Circular-phase Gamma v1.6 does not advance. Gamma remains an open, noncanonical architecture class. The laboratory preserves the negative result, repairs the cross-runtime fingerprint instrument prospectively, and returns to mechanism localization before nominating another implementation.

END OF ADDENDUM D - STAGE 2 WORKING REVISION 7

ADDENDUM E

GAMMA v1.7 MEANING-PLACEMENT RESULT

Thirty-seed context-consequence factorial, independent replication, and Version A ruling

E.1 ARCHITECTURE AND EVIDENCE BOUNDARY

Gamma v1.6 already stored a signed 23-channel consequence vector. v1.7 added a 98-component outer-product tensor binding seven families of pre-event target context to fourteen signed-mean and RMS consequence-type components. This object is a structural proxy for meaning; it is not moral value, phenomenological experience, or consciousness.

- Four matched arms: consequence only, meaning stored at write, meaning reconstructed at read, and dual meaning.
- The canonical node and engine remained untouched; no Gamma output entered state dynamics or the write gate.
- Bobby B ran 30 fresh preregistered seeds first; Codex C independently repeated them second.

E.2 FROZEN N=30 RESULT

Arm	Pooled RMSE	vs ordinary	vs shuffled	Ordinary signs
Read meaning	0.004287234450	4.4978%	8.8676%	25+ / 5-
Dual meaning	0.004332439196	3.4908%	9.2209%	25+ / 5-
Write meaning	0.004346314818	3.1817%	9.3825%	24+ / 6-
Best ordinary: type mean	0.004489145320	reference	-	-

All three meaning arms passed every directional gate, including the Bonferroni-corrected exact sign tests. All failed the two practical magnitude gates: at least 5.00% improvement over the best ordinary control and at least 10.00% over shuffled history.

DO_NOT_ADVANCE_MEANING_FACTORIAL_V1_7

E.3 ORDERING OF THE THREE MEANING ARMS

- Read meaning beat dual meaning on all 30 seeds.
- Dual meaning beat write meaning on all 30 seeds.
- Read meaning beat write meaning on all 30 seeds.
- Each context-bound meaning arm beat the no-meaning consequence-only arm on all 30 seeds.

The dual interaction was negative on all 30 seeds: no synergy under this encoding. Read meaning was strongest, missing the 5% gate by 0.502 percentage points. But the target uses present context, giving the read arm a matched context frame by construction; its ranking cannot establish that meaning generally originates at retrieval.

E.4 VERSION A ADJUDICATION

Bobby B and Codex C agreed on build, protocol, seeds, rounded world digests, pooled metrics, arm statistics, ordering, selected arm, and verdict. Their reported hashes differed because per-arm payload fingerprints were computed from raw JavaScript floating-point prediction arrays. Fifty-five numerical differences were observed; the maximum absolute difference was 7.771561172376096e-16. Every difference vanished at twelve decimals.

Adjudicated role-neutral scientific SHA-256

f88d83a18fe17cbf7e8399260d205242a3f52618abae0ca0e5016c122601e9db

Version A classifies the discrepancy as a fingerprint-instrumentation defect, not a scientific replication failure. Independent replication passes. The frozen source and raw results remain unchanged; future arm-payload fingerprints must hash twelve-decimal canonical arrays.

E.5 WHAT THE RESULT ESTABLISHES

- Binding consequence type to node context carried reproducible predictive information.
- Reconstructing meaning from present context consistently outperformed preserving the original write-context binding for a target defined using that same present context.
- Correct historical pairing mattered, but event type alone recovered most of the available signal.
- The fixed dual interpolation law diluted rather than amplified the read-meaning advantage.

E.6 WHAT THE RESULT DOES NOT ESTABLISH

- It does not prove that meaning exists only at retrieval.
- It does not eliminate stored meaning, dual-process meaning, or Gamma as architecture classes.
- It does not justify relaxing the frozen 5% or 10% thresholds after observing a near miss.
- It does not authorize causal installation in the node.

E.7 NEXT QUESTION

The next diagnostic should determine why event-type mean captured so much of the target, how much of the read arm's advantage follows algebraically from target construction, and whether any residual advantage remains after matched-context controls. Required decompositions include context family, consequence family, event class, actor-target direction, kernel mass, and archive support. The v1.7 verdict remains frozen during that work.

A future candidate should be nominated only if the diagnostic identifies a specific source of lost relational information. Reusing the current 30 seeds is permissible for exploratory localization but not for a new confirmatory claim.

E.8 STAGE 2 DECISION

No v1.7 arm advances. Read-time reconstruction becomes the leading mechanistic clue, not canonical anatomy. The laboratory preserves the near-threshold result, the negative dual interaction, and the architecture boundary without moving the goalposts.

ADDENDUM F

GAMMA v1.8 LEDGER/LENS TANDEM RESULT

Thirty-seed hard-selector test, independent replication, diagnostic interpretation, and Version A ruling

F.1 WHY v1.8 WAS RUN

v1.7's Dual arm was a convex interpolation between write-context and read-context payloads. It averaged two interpretations; it did not establish two independently stateful processes operating in tandem. v1.8 tested Bobby's proposed distinction: Gamma as a ledger of settled causal/action history and provisional Lambda as a lens that tracks which interpretation has recently predicted each consequence channel more accurately.

- Gamma Ledger: v1.7 write-context prediction, aligned into the present scoring frame.
- Lambda Lens: v1.7 read-context prediction plus separately serialized reliability history.
- Legacy Dual: the v1.7 convex blend retained as a dilution control.
- Tandem Selector: winner-take-channel routing by event kind and consequence channel; no averaging.

Lambda is provisional nomenclature. This experiment tested reliability routing, not a complete independent perception archive, neurodivergence, consciousness, or a universal cognitive law.

F.2 FROZEN FIELD AND PRIMARY ENDPOINT

The field retained 6 nodes, 30 warmup ticks, 360 acquisition events, 180 held-out test events per seed, horizon 10, archive capacity 48, circular phase geometry, and strengths 0.12, 0.20, and 0.28. Thirty new seeds were executed by Bobby B first and Codex C second.

The primary score was macro event-stratified RMSE, giving all six event kinds equal weight. This corrected the micro-averaging problem identified after v1.7, where RECOVERY_SHOCK accounted for approximately 85% of baseline squared error.

$$\text{RMSE}_{\text{macro},s} = \sqrt{\frac{1}{K} \sum_k \text{RMSE}(s,k)^2}$$

Advancement required at least 5% improvement over event type plus strength, both single operators, and Legacy Dual; no event-kind degradation above 10% relative to the better single operator; four favorable exact sign tests at Bonferroni alpha 0.0125; and B/C hash agreement.

F.3 FINAL N=30 RESULT

Arm	Macro RMSE	Tandem vs arm
Event type + strength	0.0013193867	-221.1545%
Gamma Ledger	0.0042353852	-0.0445%
Lambda Lens	0.0042350839	-0.0516%
Legacy Dual	0.0042345890	-0.0633%
Tandem Selector	0.0042372697	reference

DO_NOT_ADVANCE_TANDEM_V1_8

F.4 GATES AND SEEDWISE DIRECTION

Gate	Result
At least 5% vs ordinary baseline	FAIL
At least 5% vs both single operators	FAIL
At least 5% vs Legacy Dual	FAIL
Worst event-kind degradation at most 10%	PASS (0.4755%)
Four Bonferroni sign tests	FAIL
Independent B/C replication	PASS

Tandem lost to event type plus strength on all 30 seeds. Exact two-sided sign-test $p = 1.862645149230957e-9$. Against Gamma, Lambda, and Legacy Dual, seed directions were only 14/16 or 16/14, each $p = 0.855535551905632$. Their confidence intervals crossed zero.

The selectors were not catastrophically unstable: the worst event-kind degradation relative to the better single operator was only 0.4755%. The failure was more informative than a blow-up. Tandem behaved almost exactly like the two inputs and added no practical predictive value.

F.5 INDEPENDENT REPLICATION

All three ten-seed batch hashes matched exactly across Bobby B and Codex C. Both complete role-neutral scientific payloads produced:

```
4b0d51b4f7774181bc805bc93d27b2059547412bbfcfe9c477b83d6c1dcd83bd
```

Version A therefore found no runtime discrepancy to arbitrate. Build, protocol, seeds, rounded outputs, gate states, and verdict agree.

F.6 POST-CONFIRMATORY SELECTOR AUTOPSY

Event kind	Gamma	Lambda	Ties	Lambda share
ALIGN_THETA	1,418	676	426	26.83%
DISRUPT_PHASE	1,010	1,088	422	43.17%
ERODE_MEMORY	1,370	730	420	28.97%
INTEGRATION_PULSE	1,308	783	429	31.07%
RECOVERY_SHOCK	1,334	752	434	29.84%
SUPPORT_MEMORY	1,068	1,026	426	40.71%

These are terminal reliability preferences across seeds, nodes, and channels, not counts of event-time selections. Lambda was active, especially for phase disruption and memory support, but the error differences between the two experts were extremely small. The two streams were insufficiently complementary for hard routing to help.

F.7 MECHANICAL DIAGNOSIS

DIAG-002 had already established that event type plus event strength captured most of the readily predictable consequence variation. v1.8 nevertheless asked Gamma and Lambda to predict the entire payload. Their inherited kernel models did not explicitly incorporate the dominant strength relationship. The selector could choose between the two supplied answers, but it could not manufacture information missing from both.

- The ordinary model's macro RMSE was approximately one third of every Gamma/Lambda arm.
- Gamma, Lambda, Dual, and Tandem differed by less than one tenth of one percent.
- The hard selector therefore tested routing among nearly redundant weak experts, not complementarity over unexplained consequence.
- The negative result is consistent across every seed and independently reproduced.

F.8 WHAT THE RESULT ESTABLISHES

- Naive convex blending was not the only problem: deterministic channel routing also failed.
- A separate reliability state can operate without mutating the canonical node, but existence is not usefulness.
- Event type and strength must be treated as the ordinary causal floor, not ignored by later Gamma candidates.
- Macro event-stratified scoring successfully prevented one high-magnitude event class from deciding the experiment.

F.9 WHAT IT DOES NOT ESTABLISH

- It does not eliminate Gamma, Lambda, or tandem architecture classes.
- It does not show that a ledger and perception history must share one storage substrate.
- It does not test multiplicative coupling, residual correction, hierarchical gating, or different timescales.
- It does not establish that Lambda is the correct symbol, ontology, or final node component.

F.10 NEXT FALSIFIABLE ARCHITECTURE

The next candidate should place the ordinary predictor beneath both operators and ask them to explain only held-out residual consequence:

$$T_{\text{hat}} = T_{\text{hat}}(\text{type}, \text{strength}) + \Delta_{\text{Gamma}} + \Delta_{\text{Lambda}}$$

A factorial should compare baseline alone, baseline plus Gamma residual, baseline plus Lambda residual, additive tandem residuals, and an interaction or hierarchical gate. Residual models must be cross-fitted so no event outcome influences its own prediction. Event-kind macro scoring and the B-first/C-second replication order remain mandatory.

F.11 NEW QUESTIONS

- Does Gamma explain residual path dependence after type and strength are removed?
- Does Lambda require a genuinely independent observation stream recorded at every perception tick rather than at settled events?
- Are ledger and lens useful at different timescales: slow consolidated consequence versus fast contextual interpretation?
- Do their residuals combine additively, multiplicatively, or through a higher-order gate?
- Are benefits concentrated in DISRUPT_PHASE, SUPPORT_MEMORY, and RECOVERY_SHOCK rather than universal across events?

F.12 STAGE 2 DECISION

Do not install v1.8 in the canonical node. Preserve Gamma as an experimental consequential-history class and Lambda as a provisional complementary lens hypothesis. Advance only to a freshly preregistered residual factorial after its data boundaries, cross-fitting rule, and operator-specific archives are specified.

F.13 PRESERVATION AND PUBLIC RECORD

- The frozen v1.8 core, preregistration, runners, commissioning output, B/C batches, summaries, comparison, and Version A ruling are preserved together.
- The v1.8 role-neutral scientific SHA-256 is the authoritative replication identity; generated timestamps and execution-role labels are excluded from it.
- The public record explicitly states that the canonical simulator was not modified and no Gamma or Lambda realization was promoted.
- The post-confirmatory selector autopsy is labeled exploratory and cannot retroactively alter the frozen decision.

F.14 PRECONDITIONS FOR A RESIDUAL FACTORIAL

Requirement	Required before execution
Ordinary floor	Freeze event type + strength as the baseline prediction
Gamma target	Predict only residual causal consequence after the ordinary floor
Lambda target	Specify a distinct observation/read history and its storage timescale
Data boundary	Cross-fit every learned residual; no event predicts itself
Controls	Baseline, +Gamma, +Lambda, additive tandem, interaction/gated tandem
Endpoint	Macro event-stratified RMSE with event-kind safety bounds

No residual-factorial result may be described as Gamma earning canonical node membership unless it also survives independent replication and a later causal-feedback ablation inside the actual node engine.